

2019 HKDSE Physics

Paper 2 — Elective Sections

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Editable transcription of all question text, formulae, symbols and units. Only essential original figures are retained; repeated figure content is omitted.

Section A · Astronomy and Space Science — Multiple-choice

Official answers are placed directly below each question.

1.1

An atomic nucleus is of order 10^{-14} m and a cluster of galaxies is of order 10^6 pc. The volume ratio of the nucleus to the cluster is approximately

- A. 10^{-37}
- B. 10^{-60}
- C. 10^{-74}
- D. 10^{-111}

Official answer: D

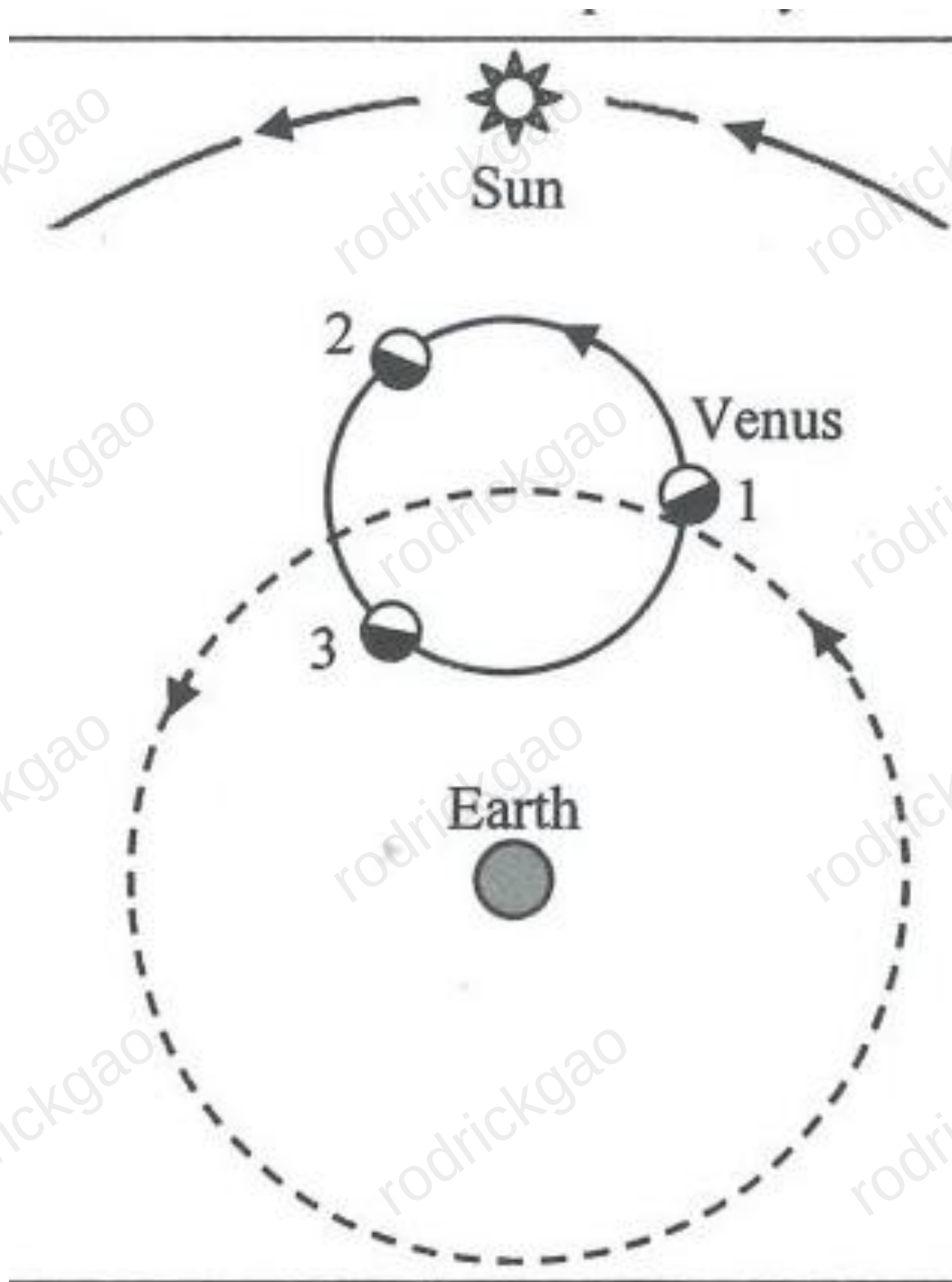
The linear-size ratio is about $10^{-14}/[10^6(3.09 \times 10^{16})] \approx 3 \times 10^{-37}$. Cubing for volume gives about 3×10^{-111} . Answer:

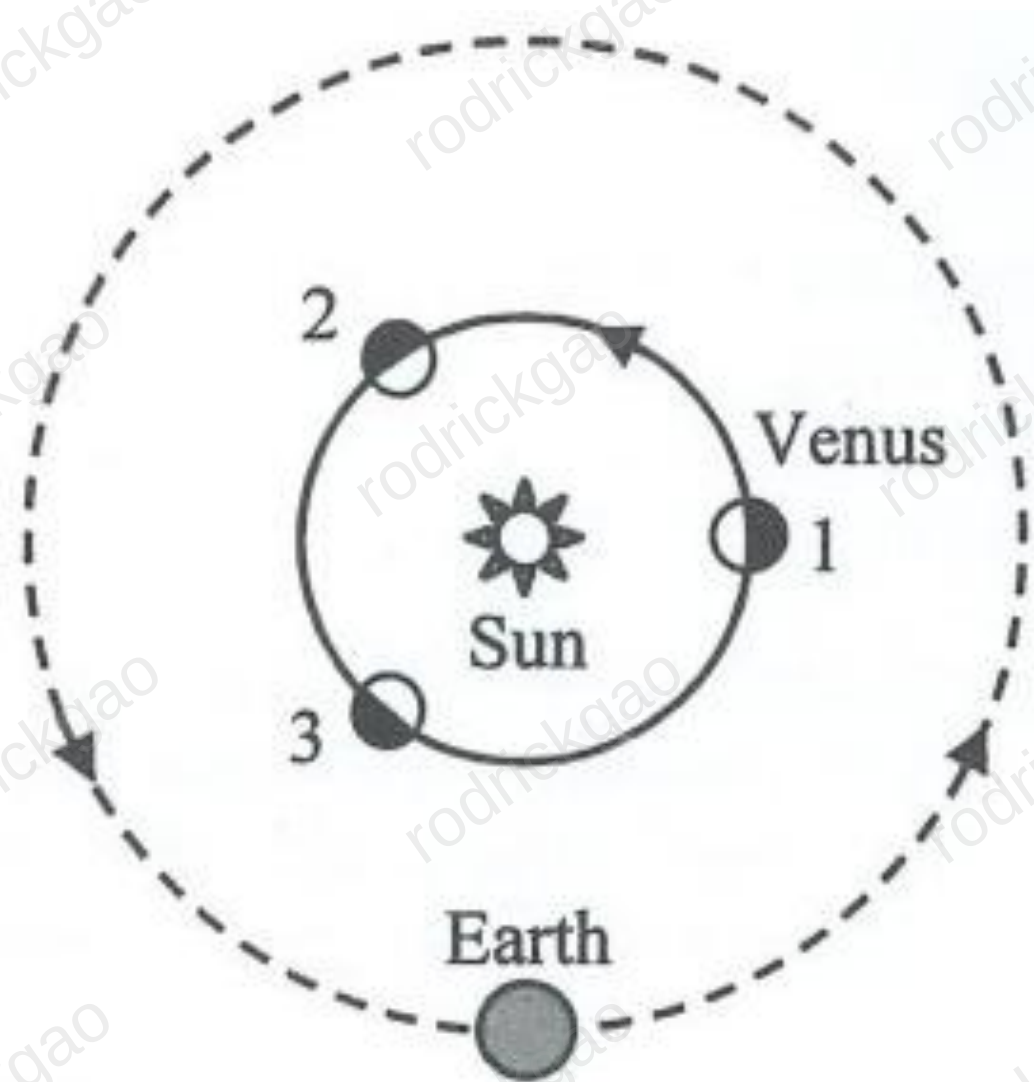
D.

1.2

The figures show three consecutive positions of Venus in geocentric and heliocentric models. Which prediction of the phases of Venus is correct?

Options A–D are shown in the original phase figure.





- | | | | | | | |
|----|---|---|---|---|---|---|
| A. |  |  |  |  |  |  |
| | 3 | 2 | 1 | 3 | 2 | 1 |
| B. |  |  |  |  |  |  |
| | 3 | 2 | 1 | 3 | 2 | 1 |
| C. |  |  |  |  |  |  |
| | 3 | 2 | 1 | 3 | 2 | 1 |
| D. |  |  |  |  |  |  |
| | 3 | 2 | 1 | 3 | 2 | 1 |
| | A | B | C | D | | |

Official answer: B

In the geocentric model Venus remains on the Earth-facing side of the Sun and cannot show the full range of phases; the heliocentric model predicts the observed changing phases. The matching pair of sequences is B.

Answer: B.

1.3

Rigel is 260 pc from the Sun. What is the angular shift on photographs taken six months apart?

- A. 0.0038"
- B. 0.0077"
- C. 130"
- D. 260"

Official answer: B

At 260 pc the annual parallax is $p = 1/260 = 0.00385''$. Photographs six months apart show the full displacement $2p = 0.00769''$. Answer: B.

1.4

Star X has luminosity 256 times the Sun and surface temperature twice the Sun. Its radius is

- A. 4 times the Sun's radius.
- B. 8 times the Sun's radius.
- C. 16 times the Sun's radius.
- D. 64 times the Sun's radius.

Official answer: A

From $L = 4\pi R^2 \sigma T^4$, $256 = (R_X/R_\odot)^2 (2)^4$. Thus $(R_X/R_\odot)^2 = 16$ and $R_X = 4R_\odot$. Answer: A.

1.5

The H β line ($\lambda = 486.1$ nm) from a star is observed at 486.6 nm. What is the star's velocity relative to Earth?

- A. 308.3 km s⁻¹ towards Earth
- B. 308.3 km s⁻¹ away from Earth
- C. 308.6 km s⁻¹ towards Earth
- D. 308.6 km s⁻¹ away from Earth

Official answer: D

The line is red-shifted by $\Delta\lambda = 0.5$ nm, so the star is receding. $v/c \approx \Delta\lambda/\lambda = 0.5/486.1$ gives $v \approx 308.6$ km s⁻¹ away from Earth. Answer: D.

1.6

Which sequence gives spectral classes in increasing surface temperature?

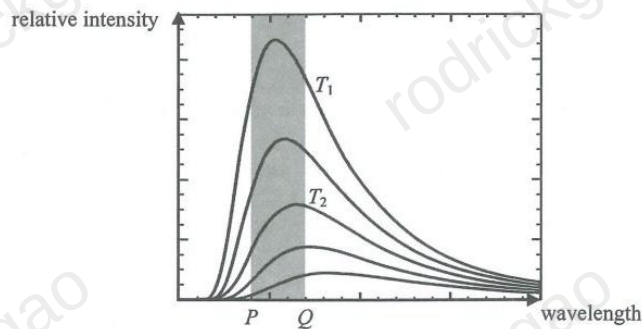
- A. KGFAO
- B. OKGFA
- C. AFGKO
- D. OAFGK

Official answer: A

Surface temperature increases along the listed spectral classes $K \rightarrow G \rightarrow F \rightarrow A \rightarrow O$. This is option A.
Answer: A.

1.7

The graph shows radiation curves. P and Q are the lower and upper limits of visible wavelength; one of T_1 and T_2 is the Sun's temperature. Which is correct?



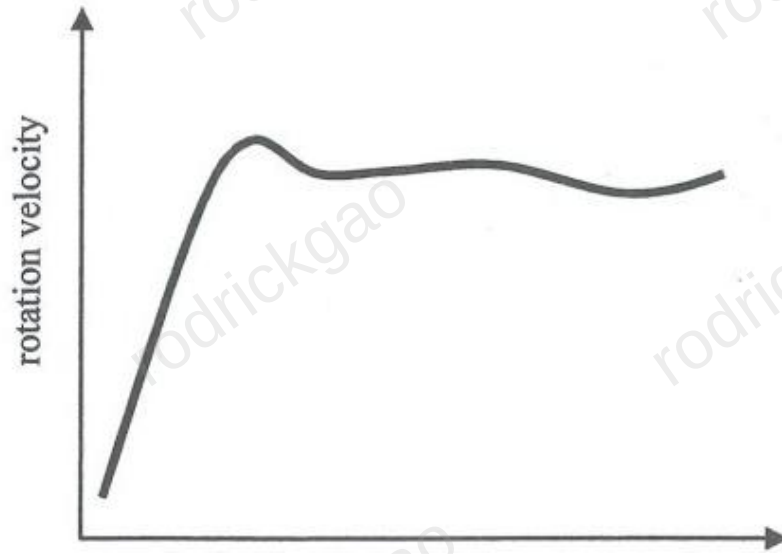
- A. P = red; Q = violet; T_1 is the Sun
- B. P = red; Q = violet; T_2 is the Sun
- C. P = violet; Q = red; T_1 is the Sun
- D. P = violet; Q = red; T_2 is the Sun

Official answer: C

P is the lower wavelength limit and is violet; Q is the upper limit and is red. The Sun's ≈ 5800 K blackbody curve is the one labelled T_1 . Answer: C.

1.8

What can be inferred about the location of dark matter from the galactic rotation curve?



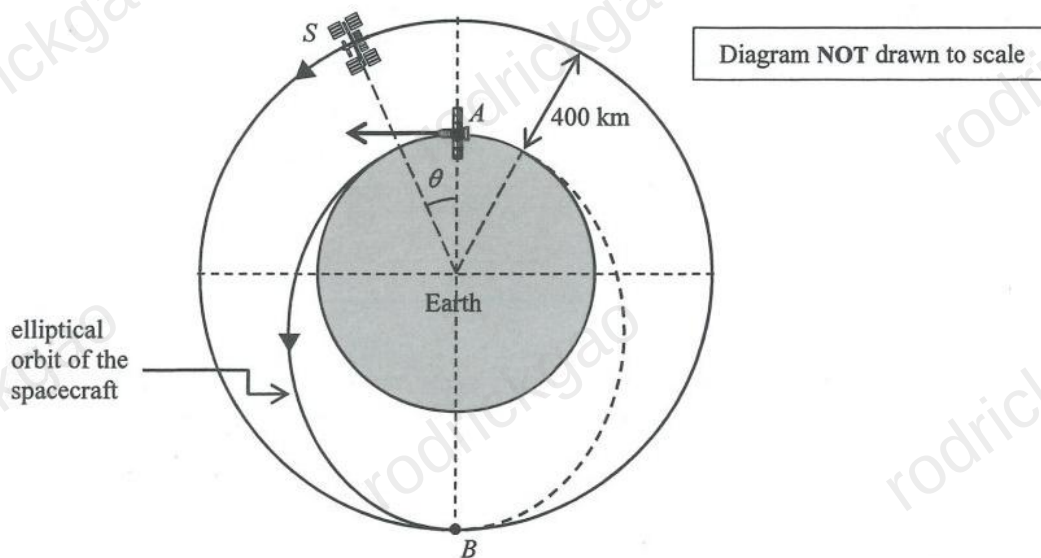
- A. It is mainly near the galactic centre.
- B. It is uniformly distributed throughout the galaxy.
- C. More of it is distributed at large distance from the centre.
- D. The curve suggests its existence but provides no information about distribution.

Official answer: C

A nearly flat rotation speed at large radius requires enclosed mass to keep increasing well beyond the bright galactic centre. This indicates more dark matter at large distance from the centre. Answer: C.

Section A · Structured Question 1

Space station S is in a circular orbit 400 km above Earth. A spacecraft is launched from A at Earth's surface at 8.02 km s^{-1} and coasts with its engine off along elliptical orbit AB, whose major axis is AB, to meet S at B. The two orbits lie in the same plane.



Given: $GM = 4 \times 10^5 \text{ km}^3 \text{ s}^{-2}$; radius of Earth = 6400 km.

- (a) (i) Using conservation of total mechanical energy, or otherwise, find the spacecraft speed v_B at B. (2 marks)
- (ii) Show that the spacecraft takes about 2663 s to travel from A to B. (2 marks)
- (iii) Explain why an astronaut in the orbiting spacecraft experiences weightlessness. (1 mark)
- (b) Station S travels at 7.67 km s^{-1} with period 5570 s.
- (i) If the spacecraft meets S exactly at B, use (a)(ii) to show that their initial angular separation θ is slightly less than 8° . (2 marks)
- (ii) A student suggests slightly adjusting the launch speed at A so that v_B exactly equals the station speed at B. Comment on feasibility. (2 marks)
- (iii) Suggest one simple way for the spacecraft at B to attain the station's speed. (1 mark)

Official marking scheme

Solution				Marks	Remarks
1. (a) (i)	$\frac{1}{2} m (v_B^2 - v_A^2) = GMm \left(\frac{1}{r_B} - \frac{1}{r_A} \right)$ $v_B^2 - 8.02^2 = 2(4 \times 10^5) \left(\frac{1}{6400 + 400} - \frac{1}{6400} \right)$ $v_B = 7.547679036 \text{ km s}^{-1} \approx 7.55 \text{ km s}^{-1}$			1M 1A 2	Correct sub. for v_A , r_A and r_B
(ii)	$T = \frac{2\pi a}{v} \quad \text{and} \quad \frac{GMm}{a^2} = \frac{mv^2}{a}$ $\therefore T^2 = \frac{4\pi^2 a^3}{GM}$ $T = 2\pi \sqrt{\frac{a^3}{GM}} \quad \text{where } a = \frac{r_A + r_B}{2} \quad \text{for elliptical orbit}$ $a = \frac{r_A + r_B}{2} = \frac{(6400) + (400 + 6400)}{2} = 6600 \text{ km}$ $T_{AB} = \frac{T}{2} = \frac{1}{2} \left\{ 2\pi \sqrt{\frac{6600^3}{4 \times 10^5}} \right\} = 2663.3962 \text{ s} \approx 2663 \text{ s}$			1M 1M 2	Correct expression/derivation for Kepler's 3 rd law Correct semi-major axis
(iii)	- The <u>gravitational force</u> acting on the astronaut is (all) used <u>for accelerating</u> the astronaut. - The astronaut and the spacecraft are under the <u>same acceleration due to gravity</u>, i.e. <u>free falling</u>. - The <u>gravitational force</u> (weight) acting on the astronaut is (all) used <u>for centripetal force</u>.			Any ONE 1A 1	NOT accept: - They have the same acceleration - The acceleration of gravity is used for centripetal force - No normal reaction to the astronaut in the spacecraft
(b) (i)	$\theta = \frac{\frac{5570}{2} - 2663}{5570} \times 360^\circ$ $= 7.8850987^\circ \approx 7.89^\circ$ Accept : $7.8^\circ \sim 7.9^\circ$			1M 1A 2	OR $\frac{2663}{5570} = \frac{180^\circ - \theta}{360^\circ}$ OR $\frac{2\pi(6800)}{7.67} \times \frac{180^\circ - \theta}{360^\circ} = 2663$ OR $\theta = \omega \Delta t = \frac{2\pi}{5570} \left(\frac{5570}{2} - 2663 \right)$ $= (1.128 \times 10^{-3} \text{ rad s}^{-1})(122 \text{ s})$
(ii)	If the launching speed at A is slightly higher (or lower), the length of the elliptical orbit's major axis will be longer (or shorter), i.e. <u>the orbit changed</u>. Thus the two orbits will no longer touch at B.			1A 1A 2	Accept: The shape of the spacecraft's orbit will be changed. Thus the two orbits cannot meet at B.
(iii)	The spacecraft has to fire its rocket briefly at B so as to boost up its speed to the required speed. (i.e. from 7.55 km s^{-1} to 7.67 km s^{-1})			1A 1	E.c.f. from a(i), if it is greater than 7.67 km s^{-1} , then the spacecraft should be slowed down by reverse firing of rocket.

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(a)(i) Conserving mechanical energy between $r_A = 6400 \text{ km}$ and $r_B = 6800 \text{ km}$ gives $v_B^2 - 8.02^2 = 2GM(1/r_B - 1/r_A)$, so $v_B = 7.55 \text{ km s}^{-1}$.

(ii) The ellipse has $a = (6400 + 6800)/2 = 6600 \text{ km}$; $T = 2\pi\sqrt{a^3/GM} = 5326 \text{ s}$, so $A \rightarrow B$ takes $T/2 = 2663 \text{ s}$.

(iii) Astronaut and spacecraft are in the same free fall, so no support force is felt.

(b)(i) The station turns $(2663/5570)360^\circ = 172.1^\circ$, leaving $\theta = 180^\circ - 172.1^\circ = 7.9^\circ$.

(ii) Changing launch speed changes the ellipse and its meeting point, so it cannot simply match the speed at B.

(iii) Fire the rocket briefly at B to raise 7.55 to 7.67 km s^{-1} .

Section B · Atomic World — Multiple-choice

2.1

There are dark lines in the Sun's spectrum because light at certain wavelengths emitted by the Sun is

- A. completely absorbed by the Sun's atmosphere.
- B. completely absorbed by Earth's atmosphere.
- C. partly absorbed by the Sun's atmosphere.
- D. partly absorbed by Earth's atmosphere.

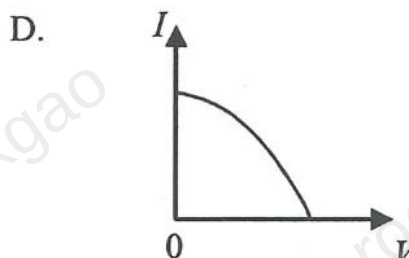
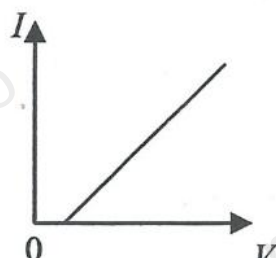
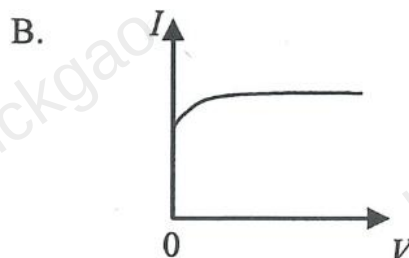
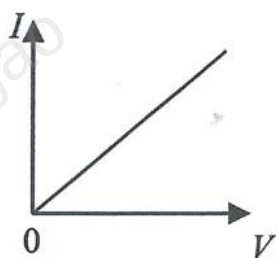
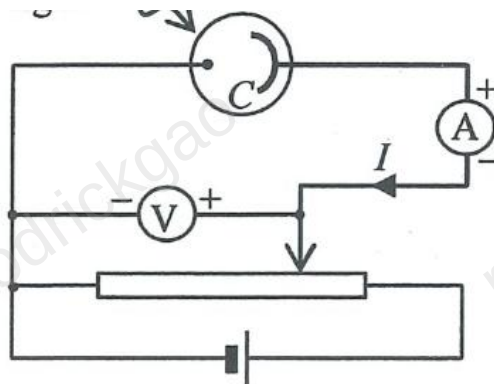
Official answer: C

Cooler gases in the Sun's atmosphere absorb part of the continuous photospheric radiation at characteristic wavelengths, producing dark absorption lines. Answer: C.

2.2

Monochromatic light of fixed intensity shines on photocell cathode C. The p.d. V is adjusted and current I measured. Which graph of I against V is correct?

Options A–D are shown in the original graph figure.



Official answer: D

With the shown polarity, increasing V increases the retarding potential. The photocurrent falls nonlinearly to zero at the stopping potential, as in graph D. Answer: D.

2.3

Gold is illuminated by electromagnetic radiation of wavelength 200 nm. Its work function is 5.30 eV. What is the maximum photoelectron kinetic energy?

- A. 0.916 eV
- B. 5.30 eV
- C. 6.22 eV
- D. 11.3 eV

Official answer: A

Photon energy is $hc/\lambda \approx 1240/200 = 6.20$ eV. Hence $K_{\max} = 6.20 - 5.30 \approx 0.90$ eV, matching 0.916 eV. Answer: A.

2.4

Which microscopes can manipulate individual atoms on conductive surfaces?

(1) optical microscope (2) scanning tunnelling microscope (3) transmission electron microscope

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

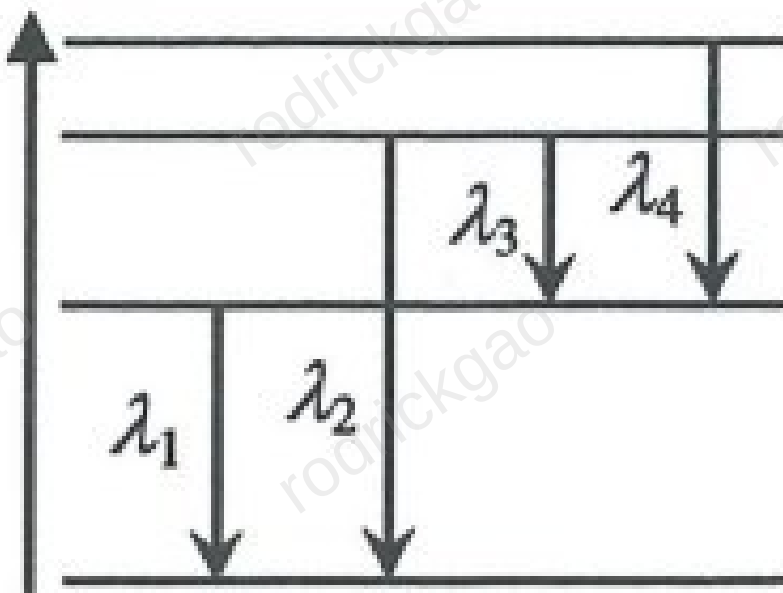
Official answer: B

A scanning tunnelling microscope can position/manipulate individual atoms on a conductive surface; optical and transmission electron microscopes do not provide this manipulation. Only (2) is correct. Answer: B.

2.5

The energy-level diagram shows four electron transitions producing emission wavelengths λ_1 , λ_2 , λ_3 and λ_4 . Which statements are correct?

- (1) $1/\lambda_3 < 1/\lambda_4$
- (2) $\lambda_1 < \lambda_2$
- (3) $\lambda_1 + \lambda_3 = \lambda_2$



- A. (1) only
- B. (2) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: A

Photon energy is hc/λ . The λ_4 transition has a larger energy gap than λ_3 , so $\lambda_3 > \lambda_4$ and $1/\lambda_3 < 1/\lambda_4$. The other wavelength relations are false; only (1) is correct. Answer: A.

2.6

A neutron of kinetic energy 1 keV has de Broglie wavelength λ . At 2 keV its wavelength is

- A. $\sqrt{2}\lambda$
- B. λ
- C. $\lambda/2$
- D. $\lambda/\sqrt{2}$

Official answer: D

For a non-relativistic particle, $\lambda = h/p = h/\sqrt{2mK}$, so $\lambda \propto 1/\sqrt{K}$. Doubling K changes the wavelength to $\lambda/\sqrt{2}$.
Answer: D.

2.7

Two red point sources 160 m from an observer can just be resolved. If replaced by violet sources, how should the observer move to just resolve them?

- A. about 280 m farther away
- B. about 120 m farther away
- C. about 120 m towards the sources
- D. about 70 m towards the sources

Official answer: B

The diffraction-limited angle is proportional to wavelength. Replacing red by violet allows the same separation to be resolved at about $160(700/400) \approx 280$ m, so the observer moves roughly 120 m farther away. Answer: B.

2.8

A metal is broken into nano-sized particles and closely packed. Which statements are correct?

- (1) Total volume is nearly unchanged but total surface area increases.
- (2) Shape changes but atomic arrangement remains unchanged.
- (3) Chemical properties change but physical properties remain unchanged.

- A. (1) only
- B. (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

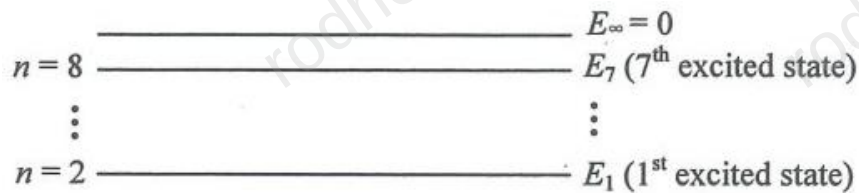
Official answer: A

Breaking the metal into nanoparticles leaves total volume nearly unchanged but greatly increases total surface area. Nanosizing can change physical as well as chemical properties, so only statement (1) is guaranteed.

Answer: A.

Section B · Structured Question 2

- (a) In Thomson's plum-pudding model, an atom is positive material with negatively charged electrons distributed throughout.
- (i) A beam of α particles is fired at gold foil. State the scattering results. (2 marks)
- (ii) Why can Thomson's model not account for these results? (1 mark)
- (b) The diagram represents hydrogen energy levels. Ground-state energy $E_0 = -13.6$ eV.



- (i) State the physical meaning of negative energy levels and the implication of an electron being at E_∞ . (2 marks)
- (ii) Find the wavelength emitted in a transition from $n = 8$ to $n = 2$. (3 marks)
- (iii) Find the minimum energy required to ionize hydrogen from its third excited state. (2 marks)

Official marking scheme

D. A (%)				O. U (%)				I. D (%)				O. A (%)			
Solution												Marks		Remarks	
2.	(a)	(i)	Most alpha particles passed (straight) through the foil, some were only slightly deflected.									1A	If the candidate does not show a difference in quantity of the particles in these two cases, at most 1A is awarded.		
			A <u>small number</u> of alpha particles were scattered at <u>large angles</u> and a few even rebounded backward.									1A			
												2			
	(a)	(ii)	Since the charge and mass of an atom in Thomson's atomic model are evenly distributed, the alpha particles <u>should not</u> be deflected (by large angles).									1A	Accept explanation for the expt. result supporting Rutherford model (Mass and charge concentrated at a small nucleus so that alpha particles can be deflected / deflected by large angles.)		
												1			
	(b)	(i)	The electron is bounded by the nucleus, i.e. energy/work must be supplied in order to remove the electron from the atom/ionize the atom. (either)									1A	NOT accept using PE instead of the total energy. NOT accept E_n is lower than E_∞ and $E_\infty = 0$ so $E_n < 0$.		
			An electron at E_∞ is not bounded by the attractive force from the nucleus, i.e. free.									1A			
		(ii)										2	Accept $\pm 13.6 \left(\frac{1}{8^2} - \frac{1}{2^2} \right)$ NOT accept $13.6 \left(\frac{1}{8^2} + \frac{1}{2^2} \right)$. Need to check substitution of h , c and ΔE for 1M. $1.60 \times 10^{-19} \text{ C}$ not required for 1M. Accept using de Broglie formula to find the wavelength: $\Delta E = mc^2$ and $\lambda = h/p = h/mc$ $\therefore \lambda = hc/\Delta E \dots$		
			(ii) $\Delta E = E_7 - E_1$ $= -13.6 \left(\frac{1}{8^2} - \frac{1}{2^2} \right)$ $= 3.1875 \text{ eV} \approx 3.1875 \times (1.60 \times 10^{-19}) \text{ J}$ $\lambda = \frac{hc}{\Delta E} = \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{3.1875 \times (1.60 \times 10^{-19})}$ $= 3.9 \times 10^{-7} \text{ m} \approx 390 \text{ nm}$									1M			
												1M			
												1A			
												3			
	(b)	(iii)	(iii) $E_3 = -\frac{13.6}{4^2} = -0.85 \text{ eV}$ Energy required $= 0 - (-0.85)$ $= 0.85 \text{ eV}$ or $1.36 \times 10^{-19} \text{ J}$									1M	Accept $E_3 = -\frac{13.6}{4^2}$ for 1M. I.E. $= E_3 = 0.85 \text{ eV} \Rightarrow$ 1M only I.E. $= \frac{13.6}{4^2} = 0.85 \text{ eV} \Rightarrow$ 1M + 1A NOT accept I.E. $= -\frac{13.6}{3^2}$ for 1M. NOT accept I.E. $= -0.85 \text{ eV}$ for 1A.		
												1A			
												2			

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(a)(i) Most α particles pass straight through or suffer small deflections; a small number scatter through large angles and a few rebound.

(ii) Thomson's diffuse positive charge and mass cannot create the intense localized field needed for large-angle scattering.

(b)(i) A negative total energy means the electron is bound; at $E_\infty = 0$ it is free/ionized.

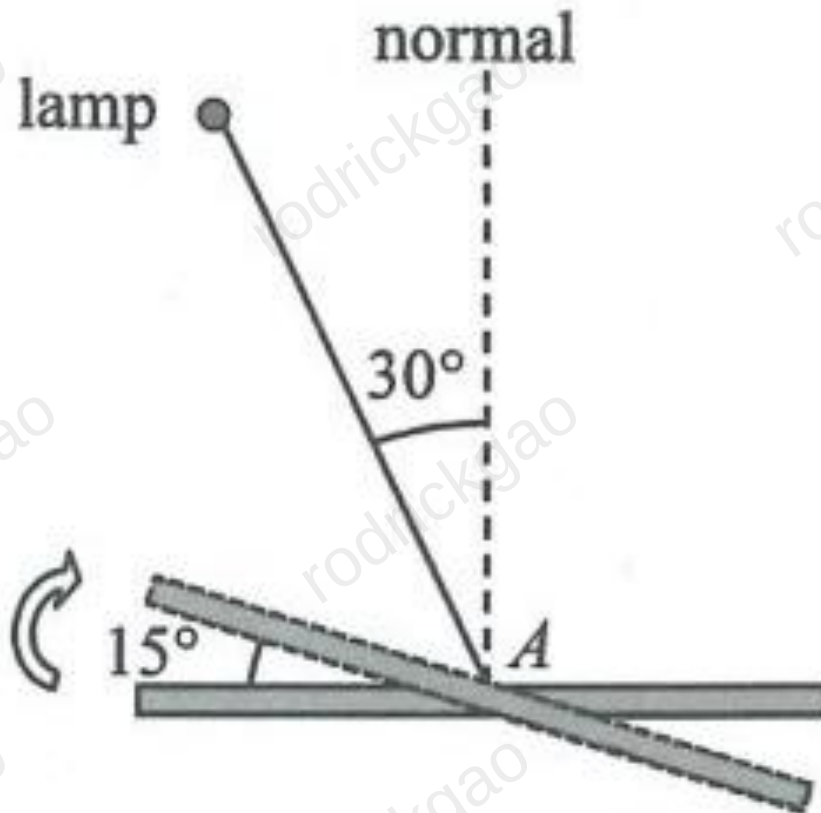
(ii) $\Delta E = 13.6(1/2^2 - 1/8^2) = 3.1875 \text{ eV}$, so $\lambda = hc/\Delta E \approx 390 \text{ nm}$.

(iii) For the third excited state $n = 4$, $E_4 = -13.6/16 = -0.85 \text{ eV}$, hence 0.85 eV is required for ionization.

Section C · Energy and Use of Energy — Multiple-choice

3.1

A horizontal book is under a lamp whose rays make 30° with the normal. Illuminance near A is 10 lx. Estimate it after the book is tilted about A by 15° as shown.



- A. 5.44 lx
- B. 8.16 lx
- C. 9.66 lx
- D. 12.2 lx

Official answer: B

Illuminance varies as $\cos\theta$. Tilting the book reduces the incidence angle from 30° to 15° , so $E_2 = 10 \cos 15^\circ / \cos 30^\circ \approx 11.2$ lx. Answer: B.

3.2

An electric-vehicle battery stores 25 kW h. Mechanical output is 12.5 kW and end-use efficiency is 80%. Estimate maximum travel duration at this output.

- A. 1.3 h
- B. 1.6 h**
- C. 2.0 h
- D. 2.5 h

Official answer: B

The useful mechanical energy from a 25 kW h battery at 80% efficiency is 20 kW h. At 12.5 kW, $t = 20/12.5 = 1.6$ h. Answer: B.

3.3

Which descriptions of a hybrid car are correct?

- (1) Motor and combustion engine can drive the car simultaneously.
- (2) It is environmentally friendly because it emits no pollutants directly.
- (3) If it cannot be wall-charged, its battery can only charge by regenerative braking during deceleration.

- A. (1) only**
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: A

A hybrid can use motor and combustion engine simultaneously. It still emits pollutants when the engine runs, and its battery may also be charged by the engine/generator, so only statement (1) is correct. Answer: A.

3.4

Air-conditioners P and Q cool identical rooms under identical conditions. Which statement is INCORRECT?

P: cooling capacity 3.2 kW, COP 2.2. Q: cooling capacity 3.0 kW, COP 1.9.

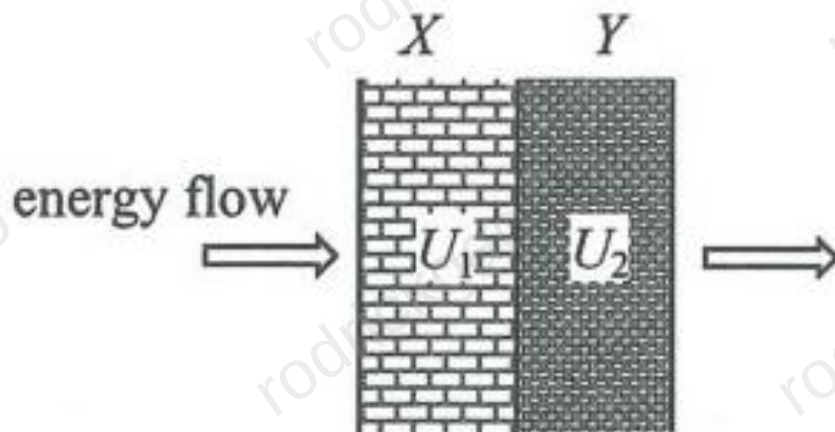
- A. The room with P cools faster.
- B. P is more energy efficient.
- C. P releases thermal energy outside at a greater rate.
- D. P consumes more power.

Official answer: D

P has greater cooling capacity and COP, so it cools faster and is more efficient. Its input power is $3.2/2.2 = 1.45$ kW, less than Q's $3.0/1.9 = 1.58$ kW; therefore 'P consumes more power' is incorrect. Answer: D.

3.5

A wall has layers X and Y of U-values U_1 and U_2 , with equal thickness and dimensions and no air gap. Its U-value is



- A. $U_1 + U_2$
- B. $\frac{1}{2}(U_1 + U_2)$
- C. $2U_1U_2/(U_1 + U_2)$
- D. $U_1U_2/(U_1 + U_2)$

Official answer: D

The two layers are thermal resistances in series: $1/U = 1/U_1 + 1/U_2$. Hence $U = U_1U_2/(U_1 + U_2)$. Answer: D.

3.6

Which arrangements can reduce a building's Overall Thermal Transfer Value (OTTV)?

- (1) Add insulation to the building envelope.
- (2) Introduce air spaces in walls.
- (3) Replace concrete walls with glass windows.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

Insulation and an air space both increase thermal resistance and reduce OTTV, while replacing concrete walls with glass generally increases heat transfer. Thus (1) and (2) are correct. Answer: A.

3.7

A wind turbine produces 800 W at wind speed 4 m s^{-1} normal to its blades. Estimate output at 6 m s^{-1} .

- A. 1200 W
- B. 1800 W
- C. 2700 W
- D. 3200 W

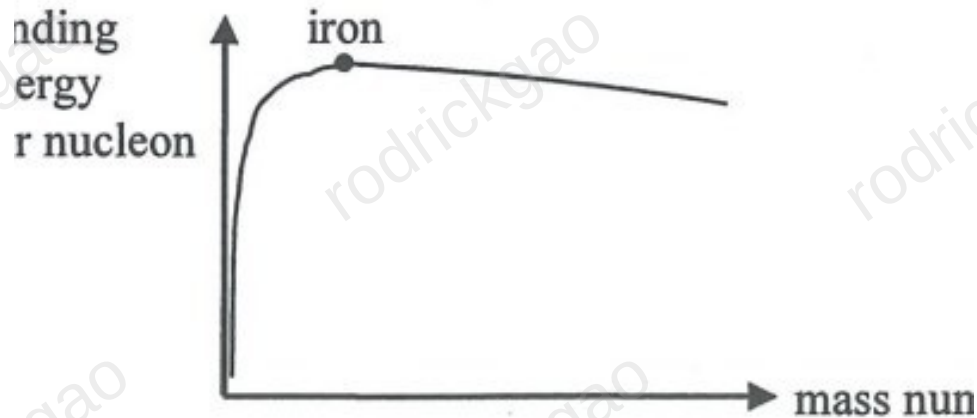
Official answer: C

For unchanged turbine area and efficiency, wind power is proportional to v^3 . $P_2 = 800(6/4)^3 = 2700 \text{ W}$. Answer: C.

3.8

Which statements can be directly inferred from the binding-energy curve?

- (1) The positive steep slope shows fusion generally releases much more energy.
- (2) The negative slope shows fission products have greater total mass than the original heavy nucleus.
- (3) Iron at the peak has the most stable nucleus.



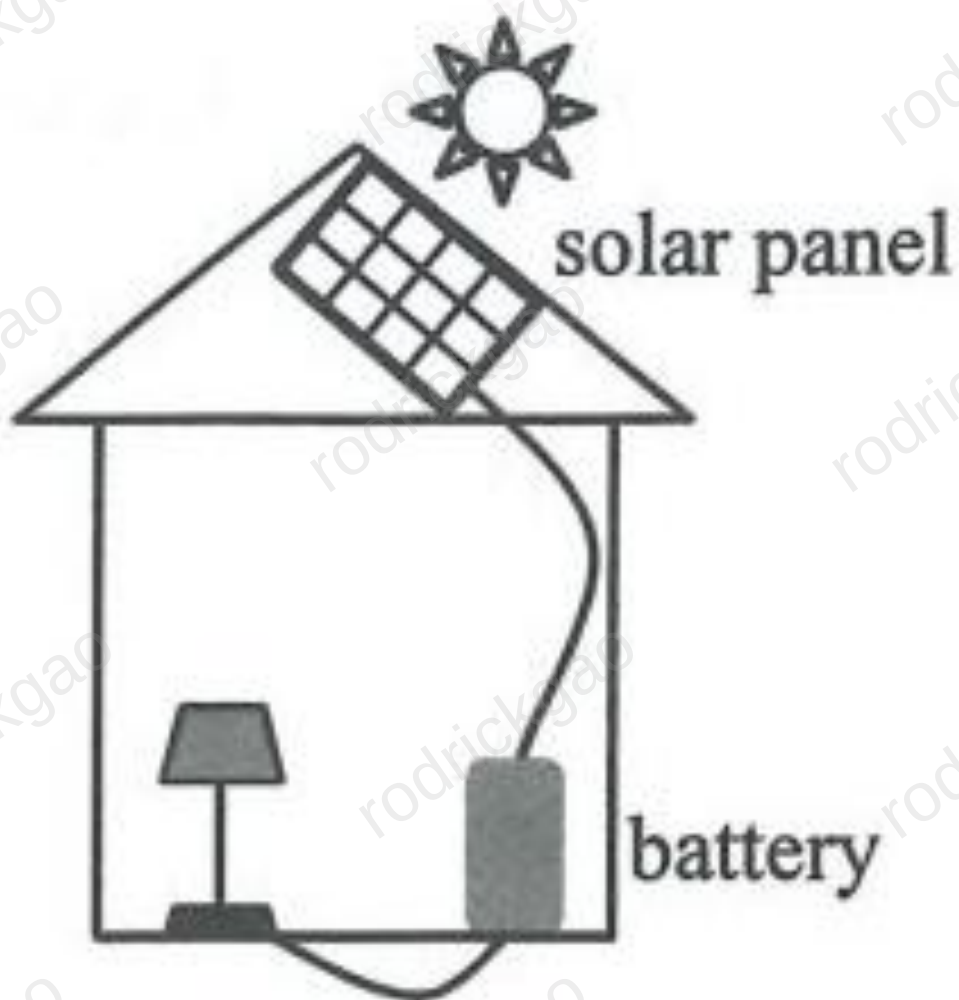
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

Fission products have lower total mass, so statement (1) is false. The steep light-nucleus side shows fusion releases more energy per nucleon, and iron at the peak is most stable; (2) and (3) are correct. Answer: C.

Section C · Structured Question 3

- (a) The Sun's total power is $3.86 \times 10^{26} \text{ W}$ and is radiated uniformly. Earth's mean distance is $1.50 \times 10^{11} \text{ m}$.
 - (i) Estimate solar-radiation power per unit area at Earth's distance. (2 marks)
 - (ii) Give one reason why the maximum received at Earth's surface normal to the Sun is only about 70% of (i). (1 mark)
- (b) The schematic shows solar energy stored in a battery through a 1.65 m^2 solar panel. The panel supplies 300 W when normal to the Sun on a sunny day; incident solar power is 1000 W m^{-2} .



(i) Describe the energy conversions during charging. (2 marks)

(ii) Find the panel efficiency. (2 marks)

(iii) A completely discharged battery is rated 100 Ah, 12 V. With 20% charging loss, find the full-charge time and state one assumption. (3 marks)

Official marking scheme

3. D (%)		4. A (%)		7. C (%)		8. C (%)	
Solution				Marks		Remarks	
3.	(a)	(i)	The radiant power coming from the Sun on unit area is given by $P_0 = \frac{P_S}{4\pi R_0^2} = \frac{3.86 \times 10^{26} \text{ W}}{4\pi (1.50 \times 10^{11})^2 \text{ m}^2}$ $= 1.365195734 \times 10^3 \text{ W m}^{-2} \approx 1365 \text{ W m}^{-2}$		1M	Note: The total (spherical) area irradiated at the Earth's orbit is $4\pi R_0^2 = 2.8274334 \times 10^{23} \text{ m}^2$ Correct sub. of P_S and R_0 Accept: $1360 - 1370 \text{ W m}^{-2}$	
					1A		
					2		
					1A		
		1	Accept: absorption / reflection / scattering by ozone layer OR some are blocked by the atmosphere				
	(b)	(i)	Solar energy $\rightarrow$ electrical energy $\rightarrow$ chemical energy 1A only for solar energy $\rightarrow$ chemical energy		1A	Accept: light energy $\rightarrow$ electric energy NOT accept: light and heat energy $\rightarrow$ electrical energy light $\rightarrow$ electricity	
					1A		
					2		
				1M	Correct sub. of input & output powers Accept: $18.0 - 18.2 \%$		
(b)	(ii)	$\eta = \frac{\text{power output}}{\text{solar power input}} \times 100\%$ $= \frac{300}{1000 \times 1.65} \times 100\%$ $= 18.1818 \% \approx 18.2 \%$		1M	Correct sub. of input & output powers Accept: $18.0 - 18.2 \%$		
				1A			
				2			
(b)	(iii)	$t = \frac{\text{total energy stored}}{\text{power input}}$ $= \frac{100 \text{ Ah} \times 12 \text{ V}}{300 \text{ W} \times 0.8}$ $= 5 \text{ hours}$		1M	1M for $\frac{100 \text{ Ah} \times 12 \text{ V}}{300 \text{ W}}$ 1A for 5 h / 300 min / 18000 s		
				1A			
				1A			
				3			
The sun rays are (always) normal to the panel Or Clear sky / not cloudy.							

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(a)(i) Solar flux at Earth's orbit is $P/(4\pi r^2) = 3.86 \times 10^{26} / [4\pi (1.50 \times 10^{11})^2] \approx 1365 \text{ W m}^{-2}$.

(ii) The atmosphere absorbs, reflects and scatters part of the radiation.

(b)(i) Solar/light energy $\rightarrow$ electrical energy $\rightarrow$ chemical energy in the battery.(ii) Efficiency = $300/[1000(1.65)] \times 100\% = 18.2\%$.(iii) Battery energy = $100 \text{ A h} \times 12 \text{ V} = 1200 \text{ W h}$; effective charging power = $0.8(300) = 240 \text{ W}$, so time = 5.0 h , assuming constant normal sunlight/clear sky.

Section D · Medical Physics — Multiple-choice**4.1**

John is long-sighted. After suitable corrective spectacles, how do near-point and far-point distances change?

- A. increased / increased
- B. increased / unchanged
- C. decreased / decreased
- D. decreased / unchanged

Official answer: C

Corrective converging spectacles for long-sightedness shift both limiting object positions closer to the eye: the near-point and far-point distances decrease. Answer: C.

4.2

An object is 20 cm in front of a concave lens. Image magnification is 0.5. Find lens power.

- A. +20 D
- B. -5 D
- C. -10 D
- D. -20 D

Official answer: B

With $u = -0.20$ m and magnification $v/u = 0.5$, $v = -0.10$ m. Thus $1/f = 1/v - 1/u = -5 \text{ m}^{-1}$, so the lens power is -5 D. Answer: B.

4.3

At 3 kHz the smallest detectable sound-level change is about 0.5 dB. This is approximately what percentage change in intensity?

- A. 12%
- B. 6%
- C. 3%
- D. 1%

Official answer: A

$0.5 \text{ dB} = 10\log_{10}(I_2/I_1)$, so $I_2/I_1 = 10^{0.05} = 1.122$. The smallest detectable intensity change is about 12%. Answer: A.

4.4

Which are non-invasive medical-imaging methods?

(1) endoscopy (2) computed tomography (CT) (3) radioactive tracers

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: D

Endoscopy enters the body and is invasive. CT scanning and radioactive-tracer imaging obtain internal information without surgical entry, so (2) and (3) are non-invasive. Answer: D.

4.5

Which statement about radiographic imaging and CT is correct?

- A. Both use different attenuation through body tissues.
- B. Radiographic X-rays are ionizing but CT uses non-ionizing radiation.
- C. CT has higher resolution because it uses gamma radiation.
- D. CT cannot image organs with cavities.

Official answer: A

Both radiography and CT rely on different attenuation of X-rays by different tissues. CT also uses ionizing X-rays, not gamma rays, and it can image organs with cavities. Answer: A.

4.6

Which statement about hot and cold spots in a radionuclide image is correct?

- A. A cold spot indicates abnormality but a hot spot does not.
- B. Both show concentration of artificial contrast medium.
- C. Both show concentration of radioactive tracer.
- D. Both show the degree of reflection from abnormal tissue.

Official answer: C

Hot and cold spots both represent relative concentration of radioactive tracer in an organ: high uptake gives a hot spot and low uptake a cold spot. Answer: C.

4.7

A tracer has effective half-life 6.9 h and biological half-life 2 days. Find physical half-life.

- A. 2.8 h
- B. 6.0 h
- C. 7.3 h
- D. 8.1 h

Official answer: D

$1/T_{\text{eff}} = 1/T_{\text{phys}} + 1/T_{\text{bio}}$. With $T_{\text{eff}} = 6.9$ h and $T_{\text{bio}} = 48$ h, $T_{\text{phys}} = [1/6.9 - 1/48]^{-1} \approx 8.1$ h. Answer: D.

4.8

A gamma-source dose rate is reduced from $24 \mu\text{Sv h}^{-1}$ to $1.5 \mu\text{Sv h}^{-1}$ by 242 mm concrete. Find the half-value thickness.

- A. 48.4 mm
- B. 60.5 mm
- C. 80.6 mm
- D. 121.0 mm

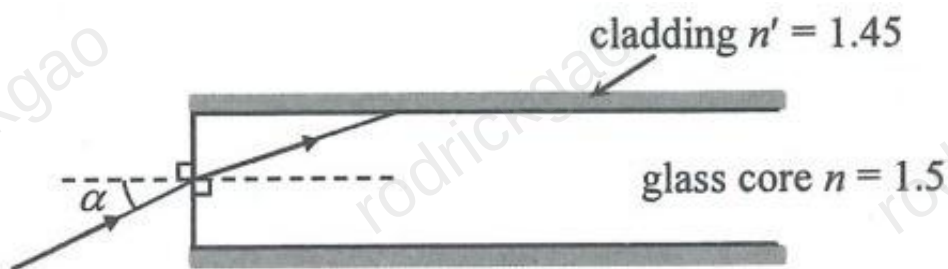
Official answer: B

The dose falls from 24 to $1.5 \mu\text{Sv h}^{-1}$, a factor of $16 = 2^4$, so 242 mm represents four half-value thicknesses.

$\text{HVT} = 242/4 = 60.5$ mm. Answer: B.

Section D · Structured Question 4

- (a) An endoscope contains optical fibres with glass core refractive index 1.50 and cladding refractive index 1.45.



- (i) Find the critical angle c at the core–cladding boundary. (1 mark)
- (ii) Explain why a ray entering the core at angle α is guided without leakage only if $\alpha < \alpha_{\text{max}}$. (2 marks)
- (iii) For a stomach ulcer, state ONE advantage and ONE disadvantage of endoscopy compared with radiographic X-rays. (2 marks)

(b) Sound-transmission data:

Tissue	Speed / m s^{-1}	Acoustic impedance / $\text{kg m}^{-2} \text{s}^{-1}$
Bone	3780	7.15×10^6
Muscle	1590	1.65×10^6
Fat	1450	1.37×10^6

(i) Estimate bone density. (1 mark)

(ii) At a muscle–bone boundary, find reflected-to-incident ultrasound intensity ratio. (2 marks)

(iii) Explain why a muscle–bone boundary is easier to distinguish than a muscle–fat boundary. (2 marks)

Official marking scheme

Solution				Marks	Remarks
4. (a) (i)	$\sin c = \frac{1.45}{1.5}$ $c = 75.2^\circ$			1A	
				1	
(ii)	For α larger than $\alpha_{\max}$, subsequently the light ray incident angle at the core-cladding boundary would be less than c , thus total internal reflection fails to occur.			1A	When $\alpha > \alpha_{\max}$, incident angle at core-cladding boundary $< c$, no total internal reflection occurs.
	For α less than $\alpha_{\max}$, subsequently the light ray incident angle at the core-cladding boundary would be greater than c , thus total internal reflection occurs.			1A	When $\alpha < \alpha_{\max}$, incident angle at core-cladding boundary $> c$, total internal reflection occurs.
	Correct description, but without mentioning core-cladding boundary			1A	Note: $\alpha_{\max} = 22.6^\circ$
				2	
(iii)	When comparing to X-rays radiographic imaging: Advantage: - direct <u>view</u> of the stomach <u>lining / inside / wall</u> - perform <u>biopsy (getting a tissue)</u> during examination if necessary - without exposure to <u>ionizing radiation</u> by X-rays Disadvantage: - requires fasting (for a few hours) prior to examination - endoscopy is an invasive procedure <u>or</u> having a risk of causing patient internal bleeding or discomfort / unwell - anesthetic may be needed - X-rays imaging is non-invasive			1A	NOT accept: Endoscope does not 'radioactive' / is clear / has higher resolution / produce real-time image / produce colour image.
				Any ONE	
				1A	Barium meal enables visualization of soft tissue using X-ray imaging. In such case, soft tissue has low contrast in X-rays radiographic image is not accepted as a correct answer.
				Any ONE	
				2	
(b) (i)	$Z_B = \rho c$ $7.15 \times 10^6 = \rho (3780)$ $\rho = 1891.534392 \text{ kg m}^{-3} \approx 1890 \text{ kg m}^{-3}$			1M/1A	Accept: $\rho = 1890 \sim 1900 \text{ kg m}^{-3}$
				1	
(ii)	$\alpha_b = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$ $= \frac{(7.15 - 1.65)^2}{(7.15 + 1.65)^2}$ $\alpha_b = \frac{I}{I_0} = 0.390625 \approx 0.391 = 39.1\%$			1M	Correct sub., the powers of Z 's can be omitted.
				1A	Accept: $\alpha_b = \left(\frac{5.5}{8.8}\right)^2 = \frac{25}{64}$ Accept: $0.39 \sim 0.391$
				2	
(iii)	The difference in acoustic impedances of a muscle-bone boundary is greater than that of a muscle-fat boundary (or vice versa), therefore giving a <u>larger intensity reflection coefficient</u> α_b ($\sim 39\%$) / <u>larger intensity reflection ratio</u> (or vice versa), <u>so more clear / easier to be distinguished.</u>			1A	Accept: $\alpha_{(\text{muscle-fat})} = 0.00859 = 0.86\%$ as a supporting statement that $\alpha_{(\text{muscle-fat})}$ is less than $\alpha_{(\text{muscle-bone})}$ The underlined statement repeats the question stem, hence NOT accepted.
				1A	

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(a)(i) $\sin c = 1.45/1.50$, giving $c = 75.2^\circ$.(ii) If the entry angle exceeds $\alpha_{\max}$, the incidence angle at the core-cladding boundary falls below c and total internal reflection fails; for $\alpha < \alpha_{\max}$, incidence exceeds c and light is guided.

(iii) Endoscopy gives a direct view/biopsy without X-ray exposure, but it is invasive and may cause discomfort, bleeding or require anaesthetic/fasting.

(b)(i) $\rho = Z/v = 7.15 \times 10^6 / 3780 = 1.89 \times 10^3 \text{ kg m}^{-3}$.(ii) $I_r/I_i = [(7.15 - 1.65)/(7.15 + 1.65)]^2 = 0.391 \approx 39.1\%$.

(iii) The muscle-bone impedance mismatch is much larger than muscle-fat, so its reflection coefficient is larger and the boundary appears clearer.